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Pedal device and electronic keyboard instrument

Abstract

A pedal device includes a base plate, a pedal supported rotatably, a rotary volume comprising a rotational shaft configured to rotate in association with the pedal, and an engagement member configured to be brought into engagement with a non-rotational portion provided around the rotational shaft of the rotary volume and with the base plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This patent application is based upon and claims the benefit of priority under 35 USC 119 of Japanese Patent Application No. 2022-066716 filed on Apr. 14, 2022, the entire disclosure of which, including the specification, claims, drawings and abstract, is incorporated herein by reference in its entirety.

BACKGROUND

Technical Field

(2) The present disclosure relates to a pedal device and an electronic keyboard instrument.

Description of the Related Art

(3) There have conventionally been disclosed pedal devices including a sensor such as a rotary volume for detecting a depressing amount of a pedal. Japanese Unexamined Patent Application Publication No. 2011-180511 (JP-A-2011-180511) discloses a sensor in which a lever portion of the sensor is pressed down via a motion transmitting portion of a pedal as a result of a depressing operation of the pedal to thereby detect a depressing amount of the pedal.

SUMMARY

(4) According to an aspect of the present disclosure, there is provided a pedal device including a base plate, a pedal supported rotatably, a rotary volume including a rotational shaft configured to rotate in association with the pedal, and an engagement member configured to be brought into engagement with a non-rotational portion provided around the rotational shaft of the rotary volume and with the base plate.

(5) According to another aspect of the present disclosure, there is provided an electronic keyboard instrument including the pedal device described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an electronic keyboard instrument including a pedal device according to an embodiment of the present disclosure;

(2) FIG. 2 is a partially sectional perspective of the pedal device according to the embodiment of the present disclosure in which, a pedal case is shown as partially being cross-sectioned;

(3) FIG. 3 is a perspective view of the pedal device according to the embodiment of the present disclosure with the pedal case and a front sheet metal member omitted from illustration;

(4) FIG. 4 is a cross-sectional view of the pedal device according to the embodiment of the present disclosure taken along a line IV-IV in FIG. 2 and viewed in a direction indicated by arrows attached to the line;

(5) FIG. 5 is an exploded perspective view of a sensor unit and a base plate of the pedal device according to the embodiment of the present disclosure as viewed from a right side;

(6) FIG. 6 is an exploded perspective view of the sensor unit and the base plate of the pedal device according to the embodiment of the present disclosure as viewed from a left side;

(7) FIG. 7 is a side view of the sensor unit of the pedal device according to the embodiment of the present disclosure as viewed from the right side; and

(8) FIG. 8 is a sectional view of the pedal device according to the embodiment of the present disclosure taken along a line VIII-VIII in FIG. 7.

DESCRIPTION OF THE EMBODIMENT

(9) Hereinafter, an embodiment of the present disclosure will be described based on drawings. An electronic keyboard instrument **10** shown in FIG. 1 includes an instrument case **14** including a base plate, a speaker, and the like which are provided in an interior thereof. The instrument case **14** includes further a full-scale (88-note) keyboard **13** including plural white keys **11** and plural black keys **12**. A leg portion **20** having four legs **21** is connected to a lower surface of the instrument case

14 (the electronic keyboard instrument **10**). The electronic keyboard instrument **10** is supported by the leg portion **20**. In the following description, a nearer side of the keyboard **13** is referred to as a front side, and an opposite side to the nearer side is referred to as a rear side. When facing the keyboard **13**, a left side is referred to as left or a left side, and a right side is referred to as right or a right side. A side of the keyboard **13** facing upwards is referred to as upper or an upper side, and a side of the keyboard **13** facing the leg portion **20** is referred to as lower or a lower side.

(10) Connecting rods **51**, that is, a left connecting rod **51** and a right connecting rod **51** are provided between the two legs **21** on the left side and the two legs **21** on the right side of the electronic keyboard instrument **10**. Although not shown, each connecting rod **51** is connected to a rear surface of the electronic keyboard instrument **10** at an upper end portion while extending downwards therefrom. The left and right connecting rods **51** are connected individually to a left side and a right side of a pedal device **50**, respectively. The pedal device **50** provided on the electronic keyboard instrument **10** includes pedals **52**, so that the electronic keyboard instrument **10** can be operated in various ways by depressing the pedals **52**. For example, a damper effect can be imparted to the electronic keyboard instrument **10**. The pedal device **50** includes a pedal case **53** which covers a base plate **161**, sensor units **160**, bearing portions **153**, and the like (which will be described later).

(11) The pedal case **53** is formed substantially into a rectangular parallelepiped box elongated in a left-right direction and including front and rear surfaces, a left and right surfaces, and an upper surface. Thus, the pedal case **53** is opened in a lower surface side. The pedal case **53** includes three opening portions **53a** formed therein. The pedal device **50** includes three pedals **52** which are aligned in a row to be positioned on a left side, a right side, and at a center (a right pedal **52a**, a left pedal **52b**, a center pedal **52c**). Each pedal **52** projects forwards (outwards) via the corresponding opening portion **52a** in the pedal case **53** while being configured to rotate.

(12) As shown in FIGS. 2 and 3, a lower base member **100** is provided in the pedal device **50**. The lower base member **100** includes a lower base member main body portion **101** elongated in the left-right direction and having a flat plate shape and three pedal facing portions **102** extending forwards from the lower base member main body portion **101** in such a manner as to face the corresponding pedals **52**. Adjustment members **103** are provided individually on a left and right pedal facing portions **102** in such a manner as to be screwed to the corresponding pedal facing portions **102** for connection thereto. The adjustment members **103** are configured to be adjusted for abutment with a setting surface. A pad **104** is provided on a central pedal facing portion **102** in such a manner as to face the setting surface. As shown in FIG. 4, cushion members **105** are provided individually on upper surfaces of the pedal facing portions **102** which are situated opposite to the adjustment members **103** and the pad **104**. The cushion members **105** are configured to be brought into abutment with lower surfaces of the corresponding pedals **52** to thereby restrict lower limit positions of the relevant pedals **52**. For example, a felt material can be used for the cushion members **105**.

(13) As shown in FIGS. 2 and 3, a lower sheet metal member **110** having an L-shaped cross section (refer to FIG. 4) is provided on a lower surface of the lower base member main body portion **101** of the lower base member **100**. The lower sheet metal member **110** includes fixing portions **111** which are provided individually at left and right end portions thereof in such a manner as to rise upright therefrom. The left and right fixing portions **111** are fixed individually to lower end portions of the left and right connecting rods **51**, respectively, with a screw member.

(14) An upper sheet metal member **120** is provided on an upper surface of the lower base member main body portion **101** of the lower base member **100**. As shown in FIG. 4, the upper sheet metal member **120** includes a lower plate portion **121**, which is disposed on the lower base member main body portion **101** into a flat plate, an upper plate portion **123** of a flat plate shape, and a connecting plate **122**, which is provided upright in a vertical direction so as to connect the lower plate portion **121** and the upper plate portion **123** together.

(15) As shown in FIG. 2, the pedal device **50** includes a front sheet metal member **130** provided in such a manner as to rise upright from an upper surface of the lower plate portion **121** with upper and lower longitudinal end portions bent rearwards. As shown in FIG. 4, the front sheet metal member **130** is fixed to the lower plate portion **121** and the lower base member main body portion **101** as a result of a lower plate portion, that is, the lower longitudinal end portion bent rearwards being fastened together therewith with a screw member. An upper plate portion of the front sheet metal member **130**, that is, the upper longitudinal end portion bent rearwards is fixed to a lower surface of an upper plate portion of the pedal case **53**. Returning to FIG. 2, the front sheet metal member **130** includes opening portions **131** formed therein in such a manner as to correspond individually to the pedals **52**. The opening portions **131** are each configured to be opened in a lower side thereof. A rib **132** is provided around a circumferential edge portion of each opening portion **131**. In FIG. 2, only the opening portion **131** (the rib **132**) is shown which corresponds to the right pedal **52a**.

(16) As shown in FIGS. 2 and 3, a restricting member **140** is provided at a proximal end portion on an upper surface side of each pedal **52**. The restricting member **140** has a first restricting portion **141** and a second restricting portion **142**. The first restricting portion **141** includes an inclined surface which is inclined towards the proximal end portion of the pedal **52** in such a manner as to cover an upper surface of the pedal **52**. The second restricting portion **142** includes an inclined surface which is disposed on the first restricting portion **141** in such a manner as to connect thereto and is inclined towards the proximal end portion of the pedal **52**. The restricting member **140** reduces a risk of the finger or the like entering the pedal device **50** via the opening portion **53a**.

(17) As shown in FIG. 3, a rectangular plate-shaped cushion member **143** is provided on an upper surface of the restricting member **140** behind the second restricting portion **142**. For example, a felt material can be used for the cushion member **143**. The cushion member **143** is brought into abutment with an upper portion of the opening portion **131** (the rib **132**) in the front sheet metal member **130** so as to restrict an upper limit position of the pedal **52** which is biased upwards by a coil spring **54**, which will be described later. Additionally, the restricting member **140** includes an arm portion **144** provided in such a manner as to project towards the sensor unit **160**, which will be described later, for engagement with a cam member **164** of the sensor unit **160** (also, refer to FIG. 4). A cam surface **144a**, which is made up of a bent planar surface portion, is provided on a lower surface of the arm portion **144** (refer to FIG. 4).

(18) An upper base member **150** is provided above the lower base member **100** via the upper sheet metal member **120**. The upper base member **150** includes spring holding portions **151** provided in such a manner as to correspond individually to portions defined above respective proximal end portions of the pedal facing portions **102** of the lower base member **100**. The spring holding portion **151** has a cylindrical shape and includes the coil spring **54** disposed in an interior thereof. The coil spring **54** is supported on a base of the spring holding portion **151** at a lower end thereof. An upper end of the coil spring **54** is connected with a spring receiving member (not shown) provided on a lower surface of the pedal **52**. The pedal **52** is biased upwards by the coil spring **54**.

(19) A proximal end portion of each spring holding portion **151** on the upper base member **150** connects with an upper base member main body portion **152**. The upper base member main body portion **152** is provided into a plate which is elongated in the left-right direction. Bearing portions **153** are provided on an upper surface of the upper base member main body portion **152** in such a manner as to correspond individually to the pedals **52**. The bearing portions **153** are configured to support rotatably rotational shafts **55** provided individually at the proximal end portions of the pedals **52**. The upper surface of the upper base member main body portion **152** is made into a recessed portion **152a** which is elongated in the left-right direction. The base plate **161** is provided in the recessed portion **152a**. The sensor units **160** are provided on an upper surface of the base plate **161**. Although this will be described later, the sensor unit **160** is such that a cam portion **164a** of the cam member **164** is brought into abutment with the cam surface **144a** of the arm portion **144**.

of the restricting member **140** for engagement, and a rotational shaft **162b** of a rotary volume **162** is interlocked with the pedal **52**.

(20) As shown in FIGS. **5** and **6**, the sensor unit **160** includes the rotary volume **162**, an engagement member **163**, the cam member **164**, a cover member **165**, and a torsion coil spring **166**. A rotary volume using a variable resistor or a rotary volume including a rotary encoder for outputting a rotational angle of the rotational shaft **162b** can be used for the rotary volume **162**. The rotary volume **162** includes three terminals **162a1** provided in such a manner as to project from a lower surface of a main body portion **162a** of a substantially rectangular parallelepiped box shape. The terminals **162a1** are inserted into terminal hole portions **161a1** formed in the base plate **161** and are then soldered. The rotary volume **162** includes the rotational shaft **162b**. The pedal device **50** includes a processor (not shown) on the base plate **161** or the like. This processor is configured to derive a depressing amount of the pedal **52** in accordance with a rotating amount of the rotational shaft **162b**. A surface cut-out portion **162b1** is provided on the rotational shaft **162b**. A portion of the rotational shaft **162b** where the surface cut-out portion **162b1** is provided has a D-shaped cross section. A substantially cylindrical non-rotational portion **162c** is provided around a proximal end portion of the rotational shaft **162b** of the rotary volume **162**. The non-rotational portion **162c** constitutes a portion which is fixed to the main body portion **162a** and never rotates together with the rotational shaft **162b**.

(21) The engagement member **163** is formed substantially into a rectangular block made up of a semi-circular upper part and a substantially flat lower surface and includes a support hole portion **163a** provided in such a manner as to penetrate the engagement member **163** in an axial direction of the rotational shaft **162b** of the rotary volume **162**. A collar portion **163b** (an abutment portion) is provided on an end side of the engagement member **163** which faces the rotary volume **162**. A boss **163c** is provided on a lower surface of the engagement member **163** in such a manner as to project downwards therefrom. The non-rotational portion **162c** of the rotary volume **162** is inserted into the support hole portion **163a** of the engagement member **163** in such a manner that an outer circumferential surface of the non-rotational portion **162c** is brought into sliding contact with an inner circumferential surface of the support hole portion **163a**. The boss **163c** of the engagement member **163** is lightly press fitted in a hole portion **161b** in the base plate **161**. Then, an abutment surface **163d**, which is the flat lower surface, is brought into abutment with the upper surface of the base plate **161**. In this way, the engagement member **163** is in engagement with the non-rotational portion **162c** of the rotary volume **162** and the base plate **161**.

(22) The cam member **164** includes a substantially cylindrical attachment portion **164b** and the cam portion **164a** provided in such a manner as to project radially outwards from the attachment portion **164b**. The cam portion **164a** is provided substantially in an axially central position of the attachment portion **164b**. A cam surface **164a1** of a convex arc-like shape is provided at a distal end portion of the cam portion **164a** in such a manner as to project to the left side. An outer circumference of the attachment portion **164b** on a side where the cam surface **164a1** projects is made into a spring disposing portion **164b1**, and the torsion coil spring **166** is disposed therearound. A projection **164c** is provided at lower portion at a left end of the attachment portion **164b** in such a manner as to project radially outwards.

(23) The cam member **164** is attached to the rotational shaft **162b** of the rotary volume **162**. As shown in FIGS. **7** and **8**, the attachment portion **164b** of the cam member **164** includes a positional aligning portion **164b2** provided in such a manner as to project from an inner circumferential surface of the attachment portion **164b**, whereby a substantially axially central portion of a substantially cylindrical interior is made into a cross-sectionally D-shaped hole. As a result, the cam member **164** is attached to the rotational shaft **162b** in such a manner that the positional aligning portion **164b2** is brought into abutment with the surface cut-out portion **162b1** of the rotational shaft **162b**. In this way, the cam member **164** is fixed to the rotational shaft **162b** while being restricted from rotating relatively to the rotational shaft **162b**.

(24) As shown in FIGS. 5 and 6, the cover member **165** includes a cover main body portion **165a**, which is opened downwards, and a frame portion **165b** of a frame shape, which is provided to extend leftwards from a lower end portion of the cover main body portion **165a**. The cover main body portion **165a** includes a rectangular parallelepiped box portion **165a1** and an arc-shaped portion **165a2** which are contiguous with each other. A bolt hole **165c1** and a bolt hole **165c2** are provided laterally outwards of the cover main body portion **165a** and the frame portion **165b**, respectively.

(25) The cover member **165** is fixedly screwed to the base plate **161** via the bolt holes **165c1**, **165c2** at bolt holes **161c**, **161d** in the base plate **161**, respectively. Here, as shown in FIG. 8, bosses **165d1**, **165d2** are provided on a lower surface of the cover member **165** in positions lying near inner sides of the bolt holes **165c1**, **165c2**, respectively. Then, the boss **165d1** is lightly press fitted in a hole portion **161e**, and the boss **165d2** is inserted in a slightly elongated hole portion **161f** which communicates with the bolt hole **161d** while the boss **165d2** is being lightly press fitted therein in a radial direction thereof.

(26) The cover member **165** covers most (the main body portion **162a**, the non-rotational portion **162c**, and a part of a proximal portion side of the rotational shaft **162b**) of the rotary volume **162**. In other words, the cover main body portion **165a** covers generally the main body portion **162a** of the rotary volume **162** with the rectangular parallelepiped box portion **165a1** and covers the engagement member **163** with the arc-shaped portion **165a2**.

(27) Here, a first movement restricting portion **165d** of a groove shape is provided on an inner surface of an upper portion of the rectangular parallelepiped box portion **165a1** substantially on a boundary portion between the rectangular parallelepiped box portion **165a1** and the arc-shaped portion **165a2**. An upper portion of the collar portion **163b** of the engagement member **163** is disposed in the first movement restricting portion **165d**. A left side wall surface (a right side wall surface in FIG. 8) of the first movement restricting portion **165d** faces the collar portion **163b**. The engagement member **163** is restricted from moving towards an axially distal end of the rotational shaft **162b** by an abutment of the relevant side wall surface of the first movement restricting portion **165d** with the collar portion **163b**.

(28) The frame portion **165b** is disposed in such a manner that the cam member **164** is accommodated within the frame portion **165b**. The projection **164c** of the cam member **164** faces a second movement restricting portion **165e** which constitutes a left inner wall surface of the frame portion **165b** of the cover member **165**. The cam member **164** is restricted from moving towards the axially distal end of the rotational shaft **162b** by an abutment of the second moving restricting portion **165e** with an end face of the projection **164c**.

(29) One (an upper) leg portion **166a** of the torsion coil spring **166** provided around the spring disposing portion **164b1** of the cam member **164** is locked on a spring locking portion **164d**, which is provided by a cut formed on a portion where the cam surface **164a1** is formed. The other (a lower) leg portion **166b** of the torsion coil spring **166** is locked on a spring locking portion **165f**, which is provided by a cut provided on a front side of the frame portion **165b** of the cover member **165**. As shown in FIG. 7, the cam member **164** is biased counterclockwise (in other words, upwardly) together with the rotational shaft **162b** of the rotary volume **162** by the torsion coil spring **166** as viewed from a distal end side of the rotational shaft **162b**. As a result, the cam portion **164a** (the cam surface **164a1**) of the cam member **164** is brought into abutment with the cam surface **144a** constituting a lower surface of the arm portion **144** of the restricting member **140** which is fixed to the upper surface of the pedal **52** for engagement therewith at all times. As a result, the cam member **164** rotates together with the rotational shaft **162b** of the rotary volume **162** in accordance with a depression of the pedal **52**, whereby a depressing amount of the pedal **52** is detected by the rotary volume **162**.

(30) In attaching the sensor unit **160** to the base plate **161**, firstly, the engagement member **163** is attached to the rotary volume **162** from the distal end side of the rotational shaft **162b** in such a

manner that the support hole portion **163a** is brought into engagement with the non-rotational portion **162c**. Subsequently, the cam member **164** is attached to the rotational shaft **162b** of the rotary volume **162**. Thereafter, the cover member **165** is attached to an assembly of the rotary volume **162**, the engagement member **163**, and the cam member **164**. Then, the torsion coil spring **166** is attached in place, whereby the sensor unit **160** is assembled. Thereafter, the sensor unit **160** so assembled is attached to the base plate **161**.

(31) In the pedal device **50**, the sensor unit **160** is provided for each pedal **52**. For example, while the sensor unit **160** may be provided only for the right pedal **52a** to impart a damper effect as on a real piano, by providing the sensor units **160** for the left pedal **52b** and the center pedal **52c**, as well, an unavailable function on a real piano (a function of exhibiting an effect in accordance with a depressing amount of the pedal **52**) can also be realized in addition to the function of simply detecting whether or not (ON/OFF) the pedal **52** is depressed.

(32) Thus, according to the embodiment of the present disclosure, the pedal device **50** includes the pedals **52** which are supported rotatably, the rotary volumes **162** provided on the base plate **161** and each including the rotational shaft **162b** configured to rotate in association with the pedal **52**, and the engagement members **163** each configured to be brought into engagement with the non-rotational portion **162c** around the rotational shaft **162b** of the rotary volume **162** and with the base plate **161**.

(33) As a result, since the rotary volume **162** is supported by the engagement member **163** which is fixed to the base plate **161**, the rotational shaft **162b** of the rotary volume **162** can be positioned in an ensured fashion. In particular, the inclination of the rotational shaft **162b** is restricted by the engagement member **163**. As a result, the rotational shafts **162b** are attached substantially uniformly in relation to attachment position for each pedal device **50**, whereby the player can experience a uniform operation feeling of the pedal **52** when he or she depresses the pedals **52** in the pedal device **50**.

(34) In addition, the pedal device **50** includes the processor configured to derive the depressing amount of the pedal **52** in accordance with the rotating amount of the rotational shaft **162b**. As a result, a damper effect of a damper or the like can be imparted in accordance with the depressing amount of the pedal **52** by which the pedal **52** is depressed.

(35) The engagement member **163** has the boss **163c** which can be inserted into the base plate **161**. As a result, the engagement member **163** can be fixed to the base plate **161** with the simple and easy construction.

(36) The engagement member **163** has the abutment surface **163d** which is brought into abutment with the base plate **161**. As a result, the rotational shaft **162b** can be positioned based upon a secure engagement which can be provided between the engagement member **163** and the non-rotational portion **162c**.

(37) The engagement member **163** has the support hole portion **163a** which penetrates the engagement member **163** in the axial direction of the rotational shaft **162b**. The non-rotational portion **162c** is inserted into this support hole portion **163a**. As a result, since the outer circumference of the cylindrical non-rotational portion **162c** can be supported by the engagement member **163** in an ensured fashion, the rotational shaft **162b** can be positioned with greater accuracy.

(38) The cover member **165** includes the first movement restricting portion **165d** configured to restrict the movement of the engagement member **163** by being brought into abutment with the abutment portion which is the collar portion **163b** of the engagement member **163** and is provided on the base plate **161** in such a manner as to cover the rotary volume **162**. As a result, even in the event that the pedal device **50** receives an impact, a risk is reduced that the engagement member **163** is caused to move towards the distal end of the rotational shaft **162b**, whereby the rotational shaft **162b** is continuously positioned in a stabler fashion.

(39) The cam member **164** is provided on the rotational shaft **162b** which includes the cam portion

164a configured to be brought into engagement with the arm portion **144** of the pedal **52**, and the cover member **165** has the second movement restricting portion **165e** configured to restrict the movement of the cam member **164**. As a result, even in the event that the pedal device **50** receives an impact, a risk is reduced that the cam member **164**, which is configured to rotate the rotational shaft **162b** in accordance with a depressing amount of the pedal **52**, is caused to move towards the distal end of the rotational shaft **162b**.

(40) The pedal device **50** has the torsion coil spring **166** of which the one leg portion **166a** is locked on the cam member **164** and the other leg portion **166b** is locked on the cover member **165**. As a result, since the cam portion **164a** (the cam surface **164a1**) of the cam member **164** can be brought into abutment with the cam surface **144a** of the arm portion **144** in an ensured fashion, a depressing amount of the pedal **52** can be detected with good accuracy.

(41) The electronic keyboard instrument **10** has the pedal device **50**. As a result, the electronic keyboard instrument **10** can be provided in which with the pedal device **50**, the uniform pedal operation feeling is provided when the pedals **52** are depressed. While the pedal device **50** is described as being provided in the electronic keyboard instrument **10** in the present embodiment, the pedal device **50** can also be provided in other instruments and electronic instruments.

(42) While the embodiment of the present disclosure has been described heretofore, the embodiment is presented as the example, and hence, there is no intention to limit the scope of the present invention by the embodiment. The novel embodiment can be carried out in other various forms, and various omissions, replacements and modifications can be made thereto without departing from the spirit and scope of the present invention. Those resulting embodiments and modified examples thereof are included in the scope and gist of the present invention and are also included in the scope of inventions claimed for patent under claims below and their equivalents.

Claims

1. A pedal device comprising: a base plate; a pedal supported rotatably; a rotary volume comprising a rotational shaft configured to rotate in association with the pedal; and an engagement member configured to be brought into engagement with a non-rotational portion provided around the rotational shaft of the rotary volume and with the base plate, wherein the engagement member comprises a support hole portion which penetrates the engagement member in an axial direction of the rotational shaft, and wherein the non-rotational portion is inserted into the support hole portion.
2. The pedal device according to claim 1, wherein the rotary volume is provided on the base plate.
3. The pedal device according to claim 1, comprising: a processor configured to derive a depressing amount of the pedal in accordance with a rotating amount of the rotational shaft.
4. The pedal device according to claim 1, wherein the engagement member comprises a boss configured to be inserted into the base plate.
5. The pedal device according to claim 1, wherein the engagement member comprises an abutment surface configured to be brought into abutment with the base plate.
6. The pedal device according to claim 1, comprising: a cover member comprising a first movement restricting portion configured to restrict a movement of the engagement member in an axial direction of the rotational shaft by being brought into abutment with an abutment portion of the engagement member and configured to cover the rotary volume.
7. The pedal device according to claim 1, comprising: a cam member provided on the rotational shaft, the cam member comprising a cam portion configured to be brought into engagement with an arm portion of the pedal; and a frame portion comprising a second movement restricting portion configured to restrict a movement of the cam member in an axial direction of the rotational shaft.
8. The pedal device according to claim 7, comprising: a torsion coil spring of which one leg portion is locked on the cam member, and a remaining leg portion is locked on the frame portion.
9. The pedal device according to claim 1, wherein the rotary volume comprises a rotary volume

main body portion of a substantially rectangular parallelepiped box shape, and wherein plural terminals configured to be inserted into the base plate are provided on the rotary volume main body portion in such a manner as to project from a lower surface thereof.

10. The pedal device according to claim 6, wherein the cover member covers a rotary volume main body portion of the rotary volume, the non-rotational portion, and a part of a proximal portion side of the rotational shaft of the rotary volume.

11. An electronic keyboard instrument comprising: a keyboard; and the pedal device according to claim 1.

12. A pedal device comprising: a base plate; a pedal supported rotatably; a rotary volume comprising a rotational shaft configured to rotate in association with the pedal; and an engagement member configured to be brought into engagement with a non-rotational portion provided around the rotational shaft of the rotary volume and with the base plate, wherein a cam member is provided on the rotational shaft, the cam member comprising a substantially cylindrical attachment portion and a cam portion provided in such a manner as to project radially outwards from the substantially cylindrical attachment portion, and wherein the cam portion is configured to be brought into engagement with an arm portion of the pedal.

13. The pedal device according to claim 12, wherein a radially outwardly projecting projection is provided at a lower portion of the substantially cylindrical attachment portion.

14. A pedal device comprising: a base plate; a pedal supported rotatably; a rotary volume comprising a rotational shaft configured to rotate in association with the pedal; an engagement member configured to be brought into engagement with a non-rotational portion provided around the rotational shaft of the rotary volume and with the base plate; and a cover member configured to cover a part of the rotary volume and the engagement member, wherein the cover member comprises a cover main body portion which is opened in a lower side and a frame portion configured to extend from a lower end portion of the cover main body portion in an axial direction of the rotational shaft.

15. The pedal device according to claim 14, wherein the cover member comprises a rectangular parallelepiped box portion and an arc-shaped portion which are provided contiguously with each other.

16. The pedal device according to claim 14, wherein bolt holes are provided individually on outer sides of the cover main body portion and the frame portion in the axial direction of the rotational shaft for fastening to the base plate with a screw.

17. The pedal device according to claim 16, wherein a boss is provided on a portion of a lower surface which lies near an inner side of the each of the bolt holes.

18. The pedal device according to claim 14, wherein the cover member covers a rotary volume main body portion of the rotary volume, the non-rotational portion, and a part of a proximal portion side of the rotational shaft of the rotary volume.

19. The pedal device according to claim 15, wherein the rectangular parallelepiped box portion covers a rotary volume main body portion of the rotary volume, and wherein the arc-shaped portion covers the engagement member.
